Assessment 2 Answers

Q1

Valid schedule using backtracking: D2→Morning, D1→Afternoon, D3→Night. Checks: D1 not Night ✓, D2 before D3 ✓, one doctor per shift ✓, D3 not Morning ✓, all assigned ✓.

Q2

Use BFS on a 5x5 grid with Manhattan heuristic only for guidance (BFS itself ignores heuristic). Expand nodes level by level, avoid obstacles, cost per move=1. Optimal path is the shortest valid path; total cost equals the number of moves. Since the grid/obstacles are not provided in the PDF, the exact path cannot be computed.

Q3

a) Constraints: blocked cells, limited sensing, risky zones (+2 cost), online updates.
b) Use an Online Search Agent with Uniform Cost Search to always expand the lowest-cost path while updating the map.
c) AI concepts: Intelligent agent—perceives and acts; Rationality—chooses minimum-cost safe path; Environment—partially observable, dynamic, sequential.
d) Justification: UCS with online updates minimizes total cost while adapting to newly discovered obstacles.